

			Release Notes		
Version	2026-01	Date	06.06.2026	Status	Final

On 2026-06-06 new data were released from the Thermodynamic Reference Database (THEREDA).

Major changes or additions compared to the former release are:

- $\text{Na}^+ - \text{K}^+ - \text{Mg}^{2+} - \text{Ca}^{2+} - \text{Cl}^- - \text{SO}_4^{2-} - \text{NO}_3^- - \text{H}_2\text{O(l)}$: binary and ternary systems (25 °C)
- $\text{M(III)} - \text{Na}^+ - \text{Mg}^{2+} - \text{Ca}^{2+} - \text{H}^+ - \text{Cl}^- - \text{NO}_3^- - \text{OH}^- - \text{H}_2\text{O(l)}$ ($\text{M} = \text{Am, Cm, Nd}$) (25 °C)
- $\text{Cd}^{2+} - \text{Na}^+ - \text{K}^+ - \text{Mg}^{2+} - \text{Ca}^{2+} - \text{Cl}^- - \text{SO}_4^{2-} - \text{CO}_3^{2-} - \text{H}_2\text{O(l)}$: binary and ternary systems (25 °C)
- $\text{Na}^+ - \text{Mg}^{2+} - \text{Ca}^{2+} - \text{OH}^- - \text{CO}_3^{2-} - \text{H}_2\text{O(l)}$ (polythermal, 25 °C – 100 °C)
- Molar volumes of solid phases, and alkalinity of solution master species (PHREEQC) were added.

Supported codes: PHREEQC, Geochemist's Workbench. The released data are provided additionally in a generic JSON format.

Released parameter files were rigorously tested. Test cases from earlier releases were re-run and new test cases for additional data added. Results with both codes agree with each other and with earlier releases.

This implies that for calculations for untested systems no quantitative measure for the description quality can be given and no warranty can be accepted for computational results. However, THEREDA is applied to an increasing number of published experimental data. Results from these applications can be accessed on our homepage under "Application examples".

The responsibility for this release lies with

- F. Bok, A. Richter: Helmholtz-Zentrum Dresden-Rossendorf, Institute for Resource Ecology, Dep. for Surface Processes, Germany:
- X. Gaona, C. Kiefer, C. Marquardt: Karlsruhe Institute of Technology, Institute for Nuclear Waste Disposal, Germany:
- D. Freyer, M. Pannach, J. Sohr, W. Voigt: TU Bergakademie Freiberg, Institute for Inorganic Chemistry, Germany
- D. Miron: Paul Scherrer Institute, Waste Management Laboratory, Switzerland
- H. C. Moog, S. Hagemann: Gesellschaft für Anlagen- und Reaktorsicherheit, Dep. for Repository Research, Germany

You are welcome to visit us at www.thereda.de/en

On behalf of the THEREDA management board

Helge C. Moog